

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_no_replay.
- Best TSPLIB ATSP holdout gap: atsp_no_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_162641_c
- started_at_local: 2026-05-13 16:26:41
- finished_at_local: 2026-05-13 16:35:58
- duration_hhmm: 00:09
- duration_seconds: 556.743
- seed_offset: 2000
- replicate_label: c
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.167273	0.024675	0.095974	0.841331	0.78	0.055992
atsp_random_replay	random	score_only	False	0.193105	0.008454	0.100779	0.801225	0.58	0.007943
atsp_failure_replay	failure	score_only	False	0.190811	0.009004	0.099908	0.828727	0.5875	0.013078
atsp_failure_replay_compression	failure	novelty_gate	True	0.187223	0.009004	0.098114	0.918878	0.58	0.037338

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.167273, synthetic 0.024675, combined 0.095974.

- Accepted-epoch count 2, mean accepted code novelty 0.841331, and final complexity 0.78.
- Adaptation efficiency 0.055992 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.167273 across 4 instances; family means: ft=0.206372, ftv=0.300062, p=0.00089, ry=0.161767.
- Panel synthetic_holdout mean gap 0.024675 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.064883.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1791, "family": "ftv", "name": "ftv35", "optimality_gap": 0.215886}, {"best_known_cost": 1286, "cost": 1533, "family": "ftv", "name": "ftv33", "optimality_gap": 0.192068}, {"best_known_cost": 1530, "cost": 1716, "family": "ftv", "name": "ftv38", "optimality_gap": 0.121569}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.193105, synthetic 0.008454, combined 0.100779.
- Accepted-epoch count 6, mean accepted code novelty 0.801225, and final complexity 0.58.
- Adaptation efficiency 0.007943 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193105 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004982, ry=0.109416.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190811, synthetic 0.009004, combined 0.099908.
- Accepted-epoch count 4, mean accepted code novelty 0.828727, and final complexity 0.58.
- Adaptation efficiency 0.013078 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190811 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004093, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.187223, synthetic 0.009004, combined 0.098114.
- Accepted-epoch count 2, mean accepted code novelty 0.918878, and final complexity 0.58.

- Adaptation efficiency 0.037338 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.187223 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004093, ry=0.087644.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Conservative Interpretation of Replay-Aware ATSP Benchmark Suite Results

Condition	Replay Mode	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Adaptation Efficiency	Mean Code Novelty	Mean Complexity	Notes on Transfer and Novelty
atsp_no_replay	none	**0.167273**	0.024675	**0.095974**	0.055992	0.841331	0.78	Best transfer observed; moderate complexity; high code novelty.
atsp_random_replay	random	0.193105	**0.008454**	0.100779	0.007943	0.801225	0.58	Best synthetic gap but worse transfer (TSPLIB gap increases); lower complexity and code novelty. Transfer worse than no replay.
atsp_failure_replay	failure	0.190811	0.009004	0.099908	0.013078	0.828727	0.5875	Similar transfer gap to random replay; code novelty slightly higher but transfer does not improve over no replay.
atsp_failure_replay_compression	failure + compression	0.187223	0.009004	0.098114	0.037338	0.918878	0.58	Slight improvement in transfer gap vs other replay modes except no replay; highest code novelty; lower complexity.

Key Points

- **Transfer performance (TSPLIB gap and synthetic holdout gap)**:
- The **no replay** condition shows the **best transfer performance** (lowest TSPLIB gap of 0.167273 and lowest transfer gap 0.095974).
- Conditions with **random replay**, **failure replay**, and **failure replay with compression** achieve better synthetic holdout gaps (~0.008–0.009) but consistently higher TSPLIB gaps (~0.187–0.193), indicating **poorer generalization to held-out TSPLIB instances**.
- This indicates that replay modes help reduce synthetic gaps but degrade transfer to real-world benchmarks.
- **Code novelty and complexity**:
- The **no replay** condition has relatively **high code novelty (0.8413)** and **higher complexity (0.78)**.

- Replay conditions have **lower complexity (~0.58)** and similar or higher code novelty (highest in failure replay with compression at 0.9189).
- Despite generally higher code novelty under replay (especially compression-aware replay), **transfer performance decreases or remains worse compared to no replay**.
- Thus, **code novelty does not translate to better transfer performance**.
- **Replay mode distinctions**:
- **No replay** leads to best transfer performance.
- **Random replay** reduces synthetic gap significantly but worsens transfer gap.
- **Failure replay (with or without compression)** improves over random replay slightly in transfer, but not enough to surpass no replay.
- **Compression-aware failure replay** attains highest novelty and lowest complexity but only modest transfer improvement relative to other replay modes.
- **Adaptation efficiency**:
- Highest in no replay (0.0559).
- Significantly lower in replay modes, especially random replay (0.0079).
- Failure replay with compression has intermediate adaptation efficiency (0.0373), but with no transfer advantage over no replay.

Summary

- The **best transfer results overall come from the no replay condition**, reflected by the lowest TSPLIB and transfer gaps.
- Replay conditions improve synthetic holdout gaps but **degrade or do not improve transfer to TSPLIB ATSP instances**.
- Despite increased code novelty in replay (especially failure replay with compression), there is no corresponding improvement in transfer; in fact, transfer performance worsens compared to no replay.
- Therefore, **code novelty increases under replay but does not equate to better transfer or algorithmic invention** as measured by deterministic metrics.
- Replay mode matters: **random replay yields poorest transfer, failure replays slightly better, but none surpass no replay in transfer performance**.
- Compression pressure under failure replay does not significantly improve transfer but boosts code novelty and reduces complexity.

Conclusion

When prioritizing transfer metrics (held-out TSPLIB and synthetic gap) and optimality gap:

- **No replay condition is most effective for transfer and adaptation efficiency despite its higher complexity**.
- Replay-aware strategies reduce synthetic gaps but harm transfer, indicating overfitting or replay bias.

- Code novelty is not correlated with improved transfer; thus, claims of algorithmic invention based solely on increased novelty under replay are unsupported.